

Momchil Gavrilov

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Executive Summary

Biomedical engineer with 4 years of experience researching how people interact with robotics, wearable technology, and medical devices. I combine biomechanics, physiology, and applied psychology with experimental design to understand user interaction and support product development. I enjoy working across engineering, clinical, and commercial teams to translate user research into technology that is safe, intuitive, and ready for market.

Experience

Robotics Engineering Intern

June 2026 -- Present

Control, Optimization, and Robotics Engineering (CORE) Lab

Davis, CA

- Instrumented a wheel-loader for autonomous operation, contributing to perception (SLAM) algorithm deployment, sensor fusion, hardware assembly, and mechanical design.

Human Factors & Research Engineer

Sep. 2023 -- Present

Bionic Engineering and Assistive Robotics (BEAR) Research Group

Davis, CA

- Authored 4 publications (journal and conference) on usability and human-machine interface (HMI) metrics across 3 human-robot interaction studies (8, 23, and 38 participants) [1] [2] [3] [4].
- Designed and executed a 38-participant randomized controlled study of user perception during human-robot interaction on a Kinarm robotic platform, using psychometric modeling, an A/B design, Bayesian equivalence testing, and 2 rounds of formative testing.
- Restored a non-functional EMG-controlled prosthetic hand (sensor integration, firmware, hardware modifications), then planned and moderated formative usability evaluations and a full-factorial, within-subject, 23-participant study.
- Redesigned a study protocol to cut session runtime from 4 to 2 hours, reducing participant fatigue and use-error risk while improving data quality.
- Led statistical analysis and visualization for a 100-respondent clinical survey evaluating awareness, adoption, and perceived utility of upper-extremity assessment tools across four rehabilitation professions.

Head Teaching Assistant --- Engineering Design; Physiology; Neurobiology

Sep. 2023 -- June 2026

University of California, Davis

Davis, CA

- Mentored 286+ students across 72 design projects, emphasizing human-centered design and iterative prototyping.

Research Assistant

Sep. 2022 -- Sep. 2023

Scientific Computing and Wearable Technology Lab

Hamilton, ON

- Deployed ankle-mounted IMUs, psychometric questionnaires, and force-plate testing in a 16-athlete, 8-month longitudinal study with a collegiate women's basketball team; ran 2-hour data collection sessions 3 days per week [5].

Clinical Assistant

Sep. 2018 -- Sep. 2021

Sports Medicine and Rehabilitation

Barrie, ON

- Assisted in patient-facing care of 1,000+ clients, gaining firsthand insight into clinical use environments and workflows.

Projects

The Wellspring App -- Donation Management Software for a Sacramento Non-profit (working demo)

- Conducted stakeholder interviews, translated user workflows into software requirements, and deployed a platform that replaced a paper-based workflow.

UC Davis Mobile: UX/UI Redesign -- Awarded "Most User-Centered Design"

- Led user research, usability testing, and iterative low- and high-fidelity prototyping for a campus-wide mobile application redesign.

Education

University of California, Davis

Davis, CA

M.S. Biomedical Engineering --- Graduate Certificate in Business Development

Sep. 2023 -- June 2026

McMaster University

Hamilton, ON

B.S. Kinesiology

Sep. 2019 -- April 2023

Skills

Usability Engineering: formative testing, moderation, task & protocol design, use-error informed design, HMI evaluation.

Experimental Design & Statistics: factorial, within-subject, RCT, A/B; Bayesian & inferential stats, psychometrics, regression.

Regulatory Knowledge: FDA HFE Guidance (2016 & 2026), IEC 62366-1, ISO 14971, ISO 13485 (Design Controls), IEC 60601, EU MDR.

Technical: R, MATLAB, Python, C/C++, CAD, EMG, biosignal acquisition & processing, biomechanics, Figma, Git, Linux, ROS2.